

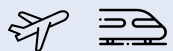
Task Construction

1 Sandbox Environment



Travel Data Sandbox

Intercity Transport



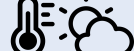
Local Mobility



POIs



Environment



40 diverse cities & dates



Environmental context

- Weather risk
- Crowd queue risk
- POI Closure
- Heat / Cold ...

2 Traveler Profile

11 diverse Traveler Profiles

Backpacker

Couples

Senior

Three-Generation Family



Student



Food-first



Friends



Businessman



3 Multi-Turn Interaction

Traveler Profiles



Base Query

We, family of five, 4-day Hong Kong trip, must visit Disneyland.



Destination Date, duration
Sandbox environment

Four Diverse Interactions

I

User-state evolution

requirements evolve across turns



Add budget Change schedule New must-visit

II

Request Resolution



5 POIs one Day

- Ask for clarification
- Tired! Conflict with Senior profiles
- Ambiguous (which date)

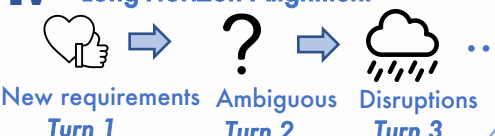
III

Environment-Driven Replanning



IV

Long-Horizon Alignment



New requirements Turn 1

Ambiguous Turn 2

Disruptions Turn 3

Agent Planning

Sandbox



Tool



- Intercity Transport
- Local Mobility
- POIs
- Environment

Decide

Clarify/NoSolution

Plan

Minute-Level Itinerary



9:00-9:30 hotel->attraction taxi
9:30-12:00 attraction
12:00-12:10 attraction->restaurant walk
12:30-13:30 meal
...

Every turn



Verifier

Interactions: new requirements, disruptions, ambiguous ...

Revision(turn-level)

- Sandbox
- Profile-Derived Rules
- Turn state
- LLM-as-judge

Evaluation

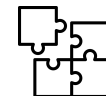
1 Response Mode

Plan

Clarification

NoSolution

2 Itinerary Feasibility



Structural Completeness



Grounded Entities



Coherent timing



Transport evidence



Cost consistency

3 Requirement Satisfaction

Hard constraints

dates, route, party size, budget, required places,, dining...

Soft Preferences (profile-conditioned)

pacing, walking burden, interest coverage...

4 Profile-Conditioned User Simulation



Simulate



Experience

- Transfer
- Visit
- Eat...

Minute-Level Activities



Physical Comfort



Schedule Comfort



Environmental Comfort



Budget Comfort



Preference Comfort

1 2 3 4 5 1 2 3 4 5 1 2 3 4 5 1 2 3 4 5 1 2 3 4 5

Stateful Multi-Turn Evaluation



Request fulfillment

Newly added or revised requirement solved?



Intent preservation

Forget or not?